1. *See supra* note 3023. Persefoni estimated that the ongoing cost is $25,000. This reflects a 44% cost reduction from its initial cost estimate, which we consider in determining the appropriate percentage reduction in subsequent calculations.
2. *See supra* note 2995. The FCA estimates initial costs to be £43,259, which is converted to dollars based on the 2022 average exchange rate. (£43,259)*(1.2369 USD/GBP) = $53,507. The ongoing costs, however, are not explicitly provided, but instead are grouped with another disclosure component. Because the initial costs make up one third of the total initial cost when combined with this other component, we assume that the same proportion holds with respect to ongoing costs. $53,507/3 = $17,836. This reflects a 67% cost reduction from its initial cost estimate, which we consider in determining the appropriate percentage reduction in subsequent calculations.
3. *See supra* note 3021.
4. *See* letter from IDFA. One unnamed company reported that it spends between 100 and 200 hours to maintain automated GHG aggregation and reporting software system for Scope 1 and Scope 2 emissions. We take the midpoint of the burden hours and convert to dollars based on $444/hr. (150 hours)*($444/hr) = $66,600.
5. *See id.* Another unnamed company reported that it spends about $100,000 to maintain its GHG measuring system, with the context suggesting that this is specific to Scope 1 emissions. Although this estimate does not include the cost of assessing Scope 2, it is nevertheless included to remain conservative in our estimation.
6. *See* letter from Energy Transfer LP. This company stated that although it already tracks Scope 1 emissions to some degree, the incremental costs to comply with the proposed rules would be approximately $7 million in 2006 dollars, which is equivalent to $10,162,035 in 2022 dollars. However, because this is only the incremental cost, it is presumably understated with respect to the full cost (i.e., incremental costs are a subset of the full cost of disclosure). It is further understated since the estimate is specific to Scope 1 emissions only, whereas we seek to estimate the costs of assessing Scope 1 and 2 emissions. Nevertheless, because this estimate is greater than the other estimates in Table 11, it is included to remain conservative in our estimation.
7. *See* letter from Soc. Corp. Gov (June 17, 2022). This company estimated that it requires roughly $300,000 annually in staff time for its Scope 1 data collection and reporting. Although this estimate does not include the cost of assessing Scope 2, it is nevertheless included to remain conservative in our estimation.

The median ongoing cost of assessing Scope 1 and 2 emissions in Table 11 is $66,600. To estimate the initial cost, we refer to two sources that reported both initial and ongoing costs to inform our assessment of the percentage reduction between the two costs. One organization's estimated costs reflect a reduction of 44 percent[3067] while another's reflect a reduction of 67 percent.[3068] We use the median (56 percent) to extrapolate the initial cost. As a result, we

[3067] *See supra* note 3023; *see also* footnote 1 in Table 11.

[3068] *See supra* note 2995; *see also* footnote 2 in Table 11.

estimate that the cost of assessing Scope 1 and 2 emissions is $151,000[3069] for the first year and $67,000 for subsequent years (rounded to the nearest $1,000).

iii. Cost estimates of assurance for Scope 1 and 2 emissions disclosures

With respect to Scope 1 and 2 emissions disclosures, the final rules require assurance at different levels (limited or reasonable) with different phase in periods depending on whether the registrant is an LAF or AF, while SRCs and EGCs are exempt.[3070] To assess the costs of assurance, we reviewed comment letters that provided relevant, quantitative cost estimates, as presented in Table 12.

The estimates displayed varying degrees of assurance "coverage" (i.e., which specific disclosures were being assured). Some commenters reported assurance costs but did not explicitly define what climate-related disclosure items were being assured.[3071] In such cases, we applied the conservative assumption that the reported assurance costs were specific to their GHG emissions disclosures only. Other estimates were specifically attributed to Scope 1 and 2 emissions,[3072] consistent with the final rules' requirements, where applicable. The majority of estimates, however, pertained to the combined assurance costs for all three scopes of emissions,[3073] which presumably overstate the assurance costs for Scope 1 and 2 emissions only. Nevertheless, we include these estimates for two reasons: first, we included them because we cannot reliably isolate the assurance costs for Scope 1 and 2 emissions only (i.e., by excluding

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3069 ($66,600)/(1-0.56) = $151,364.

3070 *See supra* section II.I.

3071 *See, e.g.,* letters from Soc. Corp. Gov (June 11, 2021 and June 17, 2022); and Persefoni.

3072 *See, e.g.,* letters from Soc. Corp. Gov (June 17, 2022); and IDFA.

3073 *See, e.g.,* letters from Soc. Corp. Gov (June 11, 2021 and June 17, 2022).

Scope 3 emissions); and second, by including costs that are overstated relative to what the final rules require, we remain conservative in our estimation.

Other commenters, however, stated that their assurance cost estimates covered both their GHG emissions and the proposed financial statement disclosures.[3074] It is likely that a significant portion of these costs is attributable to the proposed financial statement disclosures, which several commenters stated would come with high costs.[3075] We therefore did not include these estimates as they are less likely to be representative of assurance costs for Scope 1 and 2 only compared to other aggregate estimates.

The estimates also varied in the level of assurance, with most estimates equally split between either limited assurance or not specifying the level assurance. To be conservative, any estimates that did not specify the level of assurance were assumed to be limited assurance. One commenter estimated only the incremental cost of switching from limited to reasonable assurance.[3076] While we cannot infer the actual costs of either limited or reasonable assurance in this case, we nevertheless include the incremental cost because it is relatively high, allowing us to remain conservative in our estimation.

[3074] *See* letters from Soc. Corp. Gov (June 17, 2022); and Cummins.

[3075] *See* letter from ERM CVS (stating that the "fees for the [attestation for climate-related data, including GHG emissions] may be small compared to the financial audit fees" associated with the proposed rules).

[3076] *See* letter from Salesforce.

Table 12. Costs of limited assurance for GHG emissions disclosures

Commenter	**Company** (if specified)	**Limited assurance cost**
Society for Corporate Governance (June 11, 2021)	Basic Materials	$30,000 [1]
	Comm. Services	$600,000 [2]
	Health Care	$22,000 [3]
Society for Corporate Governance (June 17, 2022)	Company 1	$400,000 [4]
	Company 3	$13,000 [5]
	Company 5	$45,000 [6]
	Company 6	$15,000 [7]
	Company 7	$50,000 [8]
	Company 8	$12,500 [9]
	Company 9	$72,000 [10]
	Company 10	$15,000 [11]
	Company 11	$15,000 [12]
	Company 12	$75,000 [13]
	Company 13	$550,000 [14]
Persefoni		$82,000 [15]
International Dairy Foods Association		$62,500 [16]
Salesforce		$800,000 [17]
Median		**$50,000**

1. *See* letter from Soc. Corp. Gov (June 11, 2021). The Basic Materials company reported spending $30,000 for assurance over its Scope 1, 2, and 3 emissions without specifying the level of assurance.

2. *See id.* The Communication Services ("Comm. Services") company, which discloses Scope 1, 2, and 3 emissions (among other climate-related disclosures), reported that assurance costs are approximately $600,000 annually without specifying the coverage or level of assurance.

3. *See id.* The Health Care company, which discloses Scope 1, Scope 2, and Scope 3 (among other climate-related disclosures), reported that assurance costs are $22,000 without specifying the coverage or level of assurance.

4. *See* letter from Soc. Corp. Gov (June 17, 2022). Company 1 reported spending over $400,000 for "limited assurance from a public company accounting firm over select environmental metrics disclosed in its sustainability report, including its Scope 1, 2 (location- based and market-based), and Scope 3 (including a comparison against the base year) GHG emissions; total energy consumed; percentage grid electricity; percentage renewable energy; and water usage."

5. *See id.* Company 3 currently pays $13,000 annually for limited assurance over its Scope 1, Scope 2, and one category of Scope 3 emissions. The cost estimated may be understated given that this company believes that its current assurance may not be in compliance with the proposed rules and that costs may increase if the rule is adopted as proposed.

6. *See id.* Company 5 reported spending over $45,000 annually for "limited assurance from a professional audit firm for disclosure in its sustainability report of its Scope 1 and 2 GHG emissions and defined categories of its Scope 3 GHG emissions (exclusive of processing and use of, and end-of-life treatment for, sold products, and certain other downstream activities)."

7. *See id.* Company 6 reported spending $15,000 annually for assurance over its Scope 1 and 2 emissions and certain Scope 3 operational emissions (such as emissions associated with business travel and downstream leased assets) without specifying the level of assurance.

8. *See id.* Company 7 reported spending $50,000 annually for assurance over its Scope 1, Scope 2, and some categories of Scope 3 emissions without specifying the level of assurance.

9. *See id.* Company 8 reported spending between $10,000 and $15,000 annually for assurance over its Scope 1 and 2 emissions. We include the midpoint of this range in the table ($12,500).

10. *See id.* Company 9 reported spending $10,000 for reasonable assurance over its Scope 1, Scope 2, and some Scope 3 emissions. It also noted that another firm offered to do the same work for $180,000. To be conservative, we use this higher estimate instead. Next, we extrapolate the cost of limited assurance based on a comment letter, which states that the cost of reasonable assurance could be 2-3 times higher than limited assurance. *See* letter from Center for Climate and Energy Solutions. By taking the midpoint (2.5), we estimate the cost of limited assurance: $180,000/2.5 = $72,000.

11. *See id.* Company 10 reported spending $15,000 annually for limited assurance over its Scope 1, Scope 2, and partial Scope 3 (fuel and energy-related activities and business travel) emissions.

12. *See id.* Company 11 reported spending $15,000 annually for limited assurance over its Scope 1, Scope 2, and some Scope 3 (business travel, commuting, waste, downstream leased assets) emissions.

13. *See id.* Company 12 reported spending $30,000 for limited assurance over its Scope 1, 2, and 3 emissions. It also noted that another firm offered to do the same work for $75,000. To be conservative, we use this higher estimate instead.

14. *See id.* Company 13 reported spending $550,000 for limited assurance over its Scope 1, 2, and 3 emissions.

15. *See* ERM survey. The ERM survey indicates that 28 respondents spend an average of $82,000 for assurance/audits related to climate. According to the commenter, this "survey did not ask issuer respondents to include details of the specific level of assurance or the scope of business practices

covered, whether assurance covered all locations or all business units, or whether it consisted of limited or reasonable assurance. The costs reported by issuer respondents may include third-party assurance of Scope 1 and/or 2 GHG emissions metrics, financial metrics, or both." Although the level and coverage of assurance are unspecified, we apply the conservative assumption that the reported cost pertains to limited assurance of Scope 1 and 2 emissions.

16. *See* letter from IDFA. An unnamed, privately held company reported that it discloses Scope 1 and 2 emissions. It further states that it spends between "$50,000-$75,000 or more that is necessary to periodically hire a 3rd party consultant to review and re-validate the company's internal systems." The level of assurance is unspecified. We include the midpoint of this range in the table ($62,500).

17. *See* letter from Salesforce. This commenter did not provide actual costs of limited or reasonable assurance, but it estimated that its *incremental* cost of switching from limited to reasonable assurance over its Scope 1, 2, and 3 emissions could range from $1 to 3 million. We include this incremental cost since it serves as a lower bound for its reasonable assurance costs. We take the midpoint of this range ($2 million) and convert to limited assurance (see footnote 10 of this table): ($2 million)/2.5 = $800,000. This estimate is understated considering that it is derived from the incremental cost as opposed to actual cost.

Table 12 presents the cost estimates of limited assurance from commenters, with any adjustments or assumptions explained in the table footnotes. The median of these estimates ($50,000) is subsequently used to extrapolate the cost of reasonable assurance. One commenter stated that reasonable assurance may cost two to three times more than limited assurance, based on input from stakeholders with expertise in developing GHG inventories for companies.[3077] We use the upper end of this range and assume that reasonable assurance is three times the cost of limited assurance. As a result, we estimate that the cost of limited assurance for Scope 1 and Scope 2 emissions disclosures is $50,000, while the cost for reasonable assurance is $150,000.

Costs may vary, however, depending on the type of assurance provider. Specifically, assurance provided by a registered public accounting firm may cost more than if it were provided by a different type of service provider. However, the final rules do not require assurance to be obtained from a registered public accounting firm.[3078] Conversely, costs may be lower if a

[3077] *See* letter from Center for Climate and Energy Solutions.

[3078] One commenter suggested that most registrants will nevertheless seek assurance from registered public accounting firms to comply with the proposed rules. *See* letter from Soc. Corp. Gov. (June 17, 2022). To the extent that this is also true of the final rules, registrants may incur higher assurance costs.

registrant uses its auditor to also provide assurance over its GHG emissions disclosures rather than contracting with a different third-party. We also note that some of the companies listed in Table 12 indicated that they were unsure as to whether their current assurance practices would meet the proposed rules' requirements.[3079] We are likewise unable to make this determination without additional details on these companies' assurance practices. If these companies were to incur additional costs to meet the final rules' assurance requirements, the Commissions' compliance cost estimates may be understated in this regard. However, we believe that our conservative approach in other aspects (e.g., incorporating assurance costs that cover all three scopes of emissions instead of just Scopes 1 and 2 emissions) mitigate this concern.

iv. Estimates of reporting costs for scenario analysis and targets/goals

While the final rules do not require any registrants to undertake activities related to scenario analysis or setting targets and goals, they may require the attendant disclosures under specific circumstances,[3080] which will result in affected registrants incurring associated reporting costs. To estimate this reporting cost, we first review comment letters and other sources that inform our assessment on the costs of undertaking scenario analysis and targets or goals, then apply the assumption that 10 percent of this cost comprise the reporting cost.[3081]

Table 13 presents the relevant sources of the costs of scenario analysis.[3082] The FCA rule estimates the ongoing cost to be $40,688 for larger issuers. The BEIS rule contains ongoing cost

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3079 *See* letter from Soc. Corp. Gov (June 17, 2022).

3080 *See* sections II.G and II.D.3.

3081 The BEIS rule estimates that in the first year of compliance, the reporting cost of metrics and targets disclosure is approximately 9.4% of the cost of the "annual data gathering" activity associated with metrics and targets (*see supra* note 2996). We similarly assume that reporting costs are 10% of the cost of undertaking the associated activity.

3082 *See supra* note 3052.

estimates for two different types of scenario analysis: qualitative ($32,190) and quantitative ($79,706). Because the final rules allow for registrants to provide disclosures of either type, where applicable, we include the estimates of both. Finally, a survey indicates that the respondents' average annual expenditures is $154,000. The median of these ongoing costs is $60,197. We next extrapolate the initial cost. Some of the sources provide both the initial and ongoing cost of scenario analysis (see Table 13 footnotes), from which we determine the median percentage cost reduction (50 percent). This implies an initial cost of $120,394. Assuming that 10 percent of these costs comprise the reporting costs, we estimate that the reporting costs of scenario analysis is $12,000 in the initial year and $6,000 annually in subsequent years (rounded to the nearest $1,000).

Table 13. Costs of scenario analysis

Commenter or Source	Ongoing cost
FCA rule	$40,688 [1]
BEIS rule: qualitative scenario analysis	$32,190 [2]
BEIS rule: quantitative scenario analysis	$79,706 [3]
ERM survey	$154,000 [4]

1. *See supra* note 2995. The FCA rule estimates ongoing costs to be £32,896 for larger issuers, which is converted to dollars based on the 2022 average exchange rate. (£32,896)*(1.2369 USD/GBP) = $40,688. This reflects a 50% reduction from the initial cost estimate ($81,377), which we consider in determining the appropriate percentage reduction in subsequent calculations.
2. *See supra* note 2996. The BEIS rule estimates ongoing costs of *qualitative* scenario analysis to be £26,025, which is converted to dollars based on the 2022 average exchange rate. (£26,025)*(1.2369 USD/GBP) = $32,190. This reflects a 25% reduction from the initial cost estimate ($42,920), which we consider in determining the appropriate percentage reduction in subsequent calculations.
3. *See id.* The BEIS rule estimates ongoing costs of *quantitative* scenario analysis to be £64,440 (£52,040 for writing or quantifying scenarios and £12,400 additional cost for quality assurance and internal verification). This is converted to dollars based on the 2022 average exchange rate. (£64,440)*(1.2369 USD/GBP) = $79,706. The initial cost estimate is $240,194 (£112,400 for developing a model for conducting scenario analysis, £69,390 for writing and quantifying scenarios, and £12,400 additional cost for quality assurance and internal verification, converted to dollars based on the 2022 average exchange rate). This reflects a 67% reduction from initial to ongoing costs, which we consider in determining the appropriate percentage reduction in subsequent calculations.
4. *See* ERM survey. The ERM survey indicates that $154,000 is the average of respondents' expenditures with respect to scenario analysis, which "includes all costs to a company related to conducting assessments of the impact of climate in the short, medium, or long term using scenario analysis as well as TCFD/CDP disclosure of risks and opportunities." The survey does not include data on initial costs.

With respect to the reporting costs of targets and goals disclosure, we refer to Table 9, which presents the ongoing costs of undertaking targets and goals. The median ongoing cost of targets is $54,015. Using the median percent cost reduction from the initial year (45 percent), we extrapolate the initial cost to be $98,209. We assume 10 percent comprise the reporting

costs.[3083] Thus, we estimate that the reporting costs of targets and goals are $10,000 in the initial year of disclosure and $5,000 annually in subsequent years (rounded to the nearest $1,000).

v. Cost estimates of amendments to Regulation S-X and incremental audit fees

We reviewed comment letters that provided cost estimates pertaining to the amendments to Regulation S-X, which were often in the millions of dollars.[3084] We considered these estimates, presented in Table 14, when developing our cost estimates but made adjustments to reflect the changes made to the final rules,[3085] which we expect will substantially reduce the compliance burden compared to the proposal.

Table 14. Estimated Costs of Amendments to Regulation S-X

Commenter	Cost
Chamber of Commerce[1]	$1.5 – 2.5 million (initial) $1 – 2 million (ongoing)
Williams Companies, Inc.	"Millions of dollars"[2] (initial)
Western Energy Alliance and U.S. Oil and Gas Association	> $100 million[3] (initial)

1. The Chamber of Commerce stated that this estimate was provided by one Well-Known Seasoned Issuer it consulted regarding the proposed amendments.
2. Williams Cos. estimated the costs of implementing the proposed amendments to Regulation S-X would be in the "millions of dollars" without providing a more specific estimate.
3. Western Energy Alliance and U.S. Oil and Gas Association stated that this estimate was based on discussions with public companies that estimated costs of over $100 million for large companies when considering the need for new systems and staff training.

3083 *See supra* note 3081.

3084 *See, e.g.,* letters from API; Chamber; NRF; WEA/USOGA; and Williams Cos.

3085 *See* section II.K.